Assessment Tool - 2

Short answer

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900 channels (45 MHz / 50 KHz) per full-duplex channels since each full-duplex channel requires two 25 KHz simplex channels

80 channels (560 total channels / 7 cells)

800 channels (100 carriers × 8 time slots)

13 dB ($10 \cdot \log_{10} (80/4) = 10 \cdot \log_{10} (20) = 13$)

32 dB (-60 dBm - (-92 dBm) = 32 dB)

20 channels per cell (Total channels = 42 MHz / 300 kHz = 140 channels per cell = 140 / 7)

4 W ($P \propto d^{-4}$; doubling distance reduces power by $2^4 = 16$; 64 W/16 = 4 W)

16 times ($P_r \propto P_t \cdot d^{-2}$; doubling d reduces P_r by 4, so P_t must increase by $2^4 = 16$ for the same SNR)

9. 60 channels (4 cells × 15 channels)

10. 83.3 Hz ($f_2 = \frac{v}{\lambda} \cdot \cos\theta$; v = 27.78 m/s, λ = 0.333 m))

11. 0.625 ms (5 ms / 8)

12. 120 carriers (24 MHz / 200 KHz)

13. 6.03 dBm (8 W = 39 dBm; 39 dBm . 33 dB = 6 dBm)

14. 106.01 dB

15. 1,200 cells (6 cells × 250 cells)

16. 720,000 active cells (4000.

17. 3.75 MHz (150 × 25 KHz)

18. 1.26 MHz (29 guard bands × 40 KHz)

19. 30 handoffs (d = 3 km; distance in 1 hr = 90 km; 90/3 = 30)

20. 80,000 users (8 × 40 × 250)

21. 2,500 cells (Total area 400 km² / cell area 0.16 km², where Area $R^2 = 0.4^2 = 0.16 km^2$)

22. 200 channels (6 MHz / 30 KHz)

23. 200 Erlangs

24. 100 users

25. 720,000 users (600 users / station × 1200 stations)

26. 50 paging round (200 cells / 4 cells per round)

27. 16.67 % or 1/6 (offered traffic = 120,000 Erlangs)

28. 90,000 subscribers (600,000 × 15 %)

29. 75 % (9000 / 12,000)

30. 60 channels (Total channels = 48 MHz / 200 MHz = 240)

31. ~~250~~ 650 channels

32. 2,475 calls (2500 × 0.99)

33. −4.21 dBm

34. 6,300 users.

... km (... m/s)

... Hz ($f_d = \frac{v \cdot f}{c}$ where v = 41.66 m/s and f = 1.8 GHz)

...% or ...% (offered load = 280,000 Erlangs, capacity = 250,000; share blocked = 280,000 − 250,000 / 280,000)

5,400 users (9 cells × 600 users / cell)

240 calls (60,000 × 0.004)

150 dB (63 dBm − (−87 dBm))

375 km² (2.5 km² × 150 stations)

Each cell gets 40 channels

5. 25 paging cycles are needed

42 2 handoffs took place.

45 The SNR is 30 dB.

46 Minimum bandwidth is 37.5 Mbps

47 Total traffic generate is 250 Erlangs

48 Total bandwidth tied up is 3 MHz

49. Total supported throughput must be 400 Mbps

50. The rollout will require flowers.